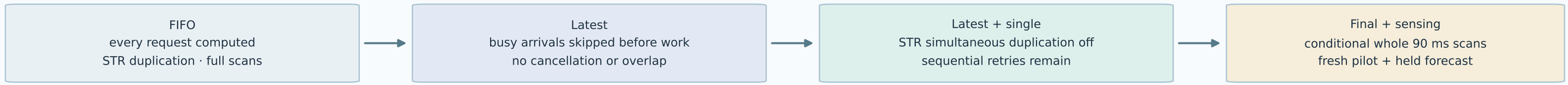
Where the enabling stack spends compute, radio time and information

Actual accepted jobs, source-time age and native whole-scan accounting; all four nested stages are applied to both fixed model families.

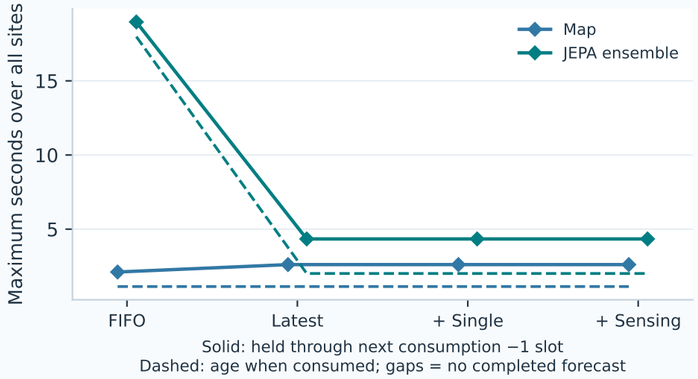
A Fixed nested intervention stack; no intermediate-arm selection



Shared: fixed RadioMap / original JEPA · actual paid FIFO service per accepted job · ordinary outage utility + all-handover admission
420 available-radio silent slots for emergency handover; full 4 s paid · current pilots / completed heard neighbors only

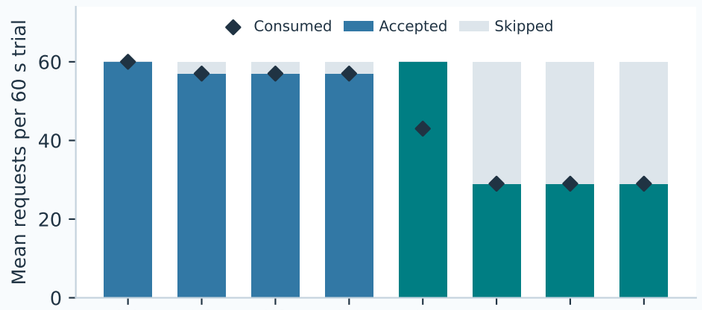
Sensing: pilot AP matches, refresh <100 ms; consumed age ≤5 s; current and best-fixed-band worst 0–5 s RSSI ≥−67 dBm.
Suppression preserves due=True; reassess after 5 s; 200 ms silence can start the whole due scan immediately. No partial scans or extra observations.

B Exact held age and consumption age



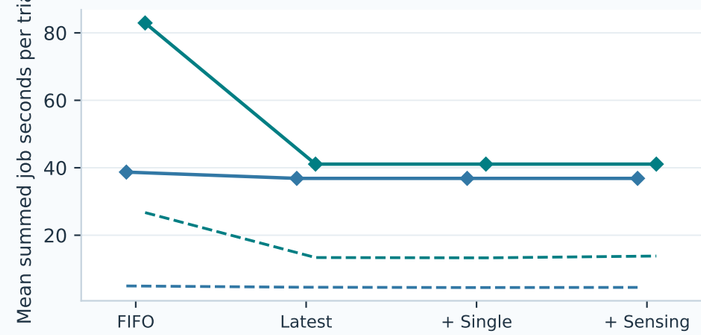
Solid: held through next consumption −1 slot
Dashed: age when consumed; gaps = no completed forecast

C Sixty arrivals: computed, skipped, consumed



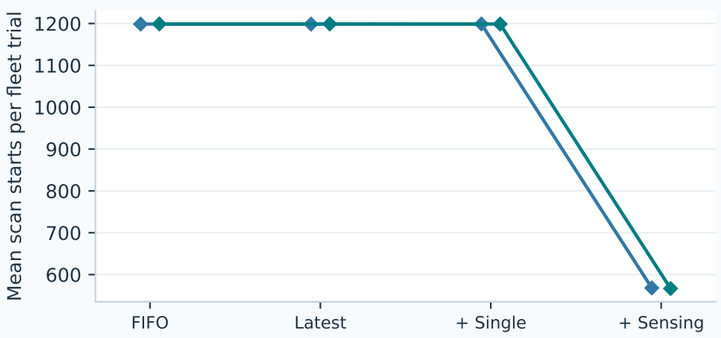
Stages 1–4 in stack order; M = Map, J = JEPA
Accepted but unconsumed jobs are still paid

D Every accepted job pays its full service



Solid: max(sealed allowance, actual issue)+0.5 s
Dashed: measured actual issue compute

E Actual whole-scan starts and suppression



Each actual start schedules a full 90 ms outage
Final suppressed assessments: Map 631.1; JEPA 632.5
Assessments are not unique skipped scans

F STR resource and original-traffic tradeoffs

Arm	STR slots (%)	Video/orig. (%)	Bulk (Mbps)	Scan charge (robot s)
M1	3.13	91.43	465.56	107.88
M2	3.13	91.45	465.52	107.88
M3	0.00	93.12	472.54	107.90
M4	0.00	93.24	472.84	51.13
J1	3.12	91.22	465.35	107.91
J2	2.96	91.30	465.22	107.91
J3	0.00	93.31	473.48	107.89
J4	0.00	93.11	474.74	51.01

STR = pending-control duplicated robot-slots / all robot-slots.
Scan charge = starts ×0.09 s; not net de-overlapped airtime.

G Paid-cost, archive and replication scope

Factual confirmation cases: 64
All source / paid / causal checks: True
Actual accepted jobs: 3,024
Skipped before computation: 816
Accepted but unconsumed: 136
Maximum initial reactive fallback: 1.96 s
Profile-exceeded cases: 1 (paid diagnostic)
Exact offline original-schedule replays: 8 / 8
Fresh-cost repetitions: 8; extra sites: 0
Fresh changed accepted source sets: 0
Fresh maximum absolute miss change: 0

Held-age extrema use every native slot from consumption through the slot before replacement; initial reactive fallback has no forecast age. One-second right-boundary snapshots are not substituted.
Whole-scan suppression changes paid sensing and delivered information. Latest rejects busy arrivals before work; no overlap or cancellation discount. Single disables simultaneous STR duplication; sequential retries remain.
Actual history-to-forecast issue work is charged; startup is separate. Admission/scoring/scan-gating CPU is not independently scheduled; common +0.5 s remains an assumption.
Fixed weights, shared procedural warehouse CAD, assumed routes/pilots, warm histories and finite episodes limit transfer. No model-objective, real-car, full-stack real-time or production claim.

Earlier failures, nulls and gates remain unchanged. New stack sites are disjoint from both prior studies.

artifacts/jepa_repair/figures/confirmation.svg

artifacts/costaware_recovery/figures/confirmation.svg